Harshad Paranjape contact@harshadparanjape.com

Education

- Doctor of Philosophy (Ph.D.) in Materials Science and Engineering *[2012-2014]*
  The Ohio State University, Columbus, OH, USA.
  Dissertation: "Modeling of Shape Memory Alloys: Phase Transformation/Plasticity Interaction at the Nano Scale and the Statistics of Variation in Pseudoelastic Performance"
- Master of Science (M.S.) in Materials Science and Engineering *[2009-2012]*
  The Ohio State University, Columbus, OH, USA.
- Bachelor of Technology (B.Tech.) in Metallurgical Engineering and Materials Science *[2005-2009]*
  Indian Institute of Technology Bombay, Mumbai, India

Research Experience

Post-doctoral Scholar at Colorado School of Mines and Visiting Post-doctoral Scholar at Northwestern University

- A study of grain-level deformation mechanisms in shape memory alloys (SMAs) using high-energy x-ray diffraction and micromechanical modeling *[2014-ongoing]*
  Characterized progression of phase transformation in single and polycrystalline SMA to reveal effects of grain-constraint, boundary conditions and test type
- A user material subroutine (UMAT) implementation for a macro-scale phenomenological model for SMA behavior *[2014-ongoing]*
  Implemented a UMAT in ABAQUS for a model coupling phase transformation and plasticity

Graduate Research at The Ohio State University

- A phase field/finite element approach to model interaction between phase transformation and plasticity in shape memory alloys (SMA) *[2012-2014]*
  Developed a new approach incorporating finite-deformation formulation, correspondence variant scale phase transformation, crystal plasticity in austenite and FEM based implementation.
- How local neighborhood affects performance of grains in polycrystalline Ni-Ti SMA *[2009-2013]*
  Developed a predictive model for grain performance based on the idea of Eshelby's inclusion stress field solution and analyzed in detail, the factors causing variation in grain performance.
- An automated method for generation of FEM mesh from confocal microscopy images of fibrous tissue scaffold *[2011-2012]*
  Initiated an effort to automatically identify individual fibers from images using MATLAB® and then generate an optimized finite element mesh to probe their mechanical properties.

Undergraduate Research Opportunity Program at Indian Institute of Technology Bombay

- Automated methods for quantifying various aspects of EBSD scans *[2008-2009]*
  Developed image processing approaches to quantify aspects (e.g. misorientation distribution at triple points, distribution of bands of Cube oriented grains) of EBSD images of deformed Al alloys.

Teaching Experience and Training

- Co-instructor for graduate Solid Mechanics class. Co-developed an "inverted classroom" methodology for the class. *[2015]*
- Volunteered to develop and incorporate MATLAB® based modules in the undergraduate computational materials science lab curriculum at The Ohio State University.
- Teaching assistant for deformation processing lab *[2012]*
  Taught use of ABAQUS® FEM software to a class of 36 students through classroom teaching, lab tutorials and exercises.
- The 2nd Summer School for Integrated Computational Materials Education, University of Michigan, Ann Arbor, MI, USA *[2012]*
  Learnt through guest lectures and exercises about incorporating computational materials science in undergraduate and graduate curriculum.

Key Research and Academic Skills

- Extensive FEM modeling experience including constitutive model development across multiple length scales and numerical implementation in ABAQUS FEM platform
- Strong foundation in programming (Python, FORTRAN, JavaScript, Matlab) and design (full stack web development, graphic design)

Selected Conference Presentations

Oral Presentation

1. Paranjape H., Bucsek A., Paul P., Park J-S., Sharma H., Dale D., Stebner A. P., Dunand D., Brinson L. C., *A Grain Scale Analysis of Phase Transformation in Shape Memory Alloys – A Coupled Synchrotron X-ray Diffraction and Micromechanical Modeling Study*, ESOMAT 2015, Antwerp, Belgium, September 13-17, 2015
2. Paranjape H., Park J-S., Sharma H., Stebner A. P., Brinson L. C., *Effect of Granular Constraints on Phase Transformation in Shape Memory Alloys – A Coupled Synchrotron X-ray Diffraction and Micromechanical Modeling Study*, Denver X-ray Conference, Denver CO, August 6, 2015
3. Paranjape H., Manchiraju S., Anderson P. M., *A Phase Field/Finite Element Model to Simulate Plasticity and Martensitic Phase Transformation in Shape Memory Alloys*, TMS Annual Meeting, San Diego CA, February 16-20, 2014
4. Paranjape H., Manchiraju S., Gao Y., Wang Y., Anderson P. M., *A Finite Element/Phase Field Approach to Study Martensitic Phase Transformation in Shape Memory Alloys*, TMS Annual Meeting, San Antonio TX, March 3-7, 2013

Poster Presentation

1. Paranjape H., Anderson P. M., *A Phase Field/Finite Element Model to Simulate Martensitic Phase Transformation in Shape Memory Alloys*, Physical Metallurgy Gordon Research Conference, Biddeford ME, July 28-August 3, 2013

Selected Publications

In Preparation

1. Paranjape, H., Anderson, P. M. *A Phase Field/Finite Element Approach to Model Coupled Phase Transformation and Plasticity in Shape Memory Alloys*, *In Preparation*

Published

2. Stebner, A. P., Paranjape H., Clausen B., Brinson L. C., & Pelton A. R. (2015). *In Situ Neutron Diffraction Studies of Large Monotonic Deformations of Superelastic Nitinol*, *Shape Memory and Superelasticity*
3. Paranjape, H., Anderson, P. M. (2014). *Texture and Grain Neighborhood Effects on Ni-Ti Shape Memory Alloy Performance*, *Modeling and Simulation in Materials Science and Engineering*.
4. Ebersole, G. C., Paranjape, H., Anderson, P. M., & Powell, H. M. (2012). Influence of hydration on fiber geometry in electrospun scaffolds. *Acta Biomaterialia*, *8*(12), 4342–4348. doi:10.1016/j.actbio.2012.07.028
5. Raveendra, S., Kanjarla, A., Paranjape, H., Mishra, S., Delannay, L., Samajdar, I., & Van Houtte, P. (2011). Strain Mode Dependence of Deformation Texture Developments: Microstructural Origin. *Metallurgical and Materials Transactions A*, 1–12
6. Raveendra, S., Paranjape, H., Mishra, S., Weiland, H., Doherty, R. D., & Samajdar, I. (2009). Relative Stability of Deformed Cube in Warm and Hot Deformed AA6022: Possible Role of Strain-Induced Boundary Migration. *Metallurgical and Materials Transactions A*, *40*, 2220–2230. doi:10.1007/s11661-009-9898-x

References

References are provided separately.